

Cheapflation Cycles

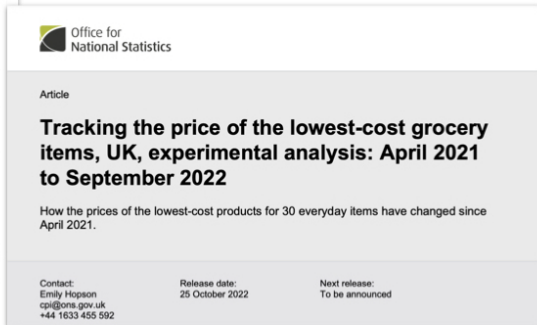
Kunal Sangani

May 2026

Disclaimer

This presentation contains my own analyses calculated (or derived) based in part on data from Nielsen Consumer LLC and marketing databases provided through the NielsenIQ Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business. The conclusions drawn from the NielsenIQ data are those of the author and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein.

Post-Pandemic “Cheapflation”



- Discussion: “supermarkets are recouping their margins on value/budget products” or “targeted fiscal stimulus increased demand at low-end.”

This Paper

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- Due to special circumstances of 2021–23 (e.g., pandemic stimulus, greedflation)?
 - **No.** The same pattern accompanies other periods in the data when input costs rose.
- Why do we see these cycles in cheapflation / inflation inequality?
 - **Simple explanation.**
 - 1 Complete pass-through in levels of input cost increases to prices.
 - 2 Same absolute price increase $\Rightarrow$ higher inflation for low-priced varieties.
 - 3 Low-income households buy lower-price varieties.
 - Explains 70% of variation in inflation inequality over time, fully accounts for heightened inflation inequality during Great Recession ('08–'11) and '21–'23.

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 - **Aggregation.** Many study inflation differences due to spending shares across categories.
 - Low-income households devote larger budget share to necessities like food, energy.
See e.g., Hobijn and Lagakos (2005), Orchard (2022), Jaravel (2024).
 - But BLS statistics miss differences in inflation across varieties **within categories**.
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 - But BLS statistics miss differences in inflation across varieties **within categories**.
 - Differential volatility, pass-through for low-income hh's understated by 70–90%.
- Welfare implications accounting for substitution, nonhomotheticity.
- Evidence that mechanism extends beyond food at home.
- Same mechanism: Other units (cities, countries) have diff. exposure to cost shocks.

Selected Related Literature

- **Inflation inequality:**

- *Using differences in spending across categories:* Michael (1979), Hagemann (1982), Garner et al. (1996), Hobijn and Lagakos (2005), Hobijn et al. (2009), Klick and Stockburger (2021, 2024), Orchard (2022), Hochmuth et al. (2022), Pallotti et al. (2023), Cavallo (2024), Jaravel (2024), Lan et al. (2024), Olivi et al. (2024), Lokshin et al. (2025).
- *Scanner data, secular trend:* Kaplan and Schulhofer-Wohl (2017), Jaravel (2019).
 - Del Canto et al. (2025): “Prior work has found that households at different income levels experience different trend inflation in consumption prices, and that this difference is driven by differences within fine product groups. [...However,] **there is no a priori reason to think inflation rates of finer product categories should be differentially responsive to short-run shocks.** We therefore limit attention to the 25 CPI groups.”
- *Scanner data, specific episodes: Great Recession:* Argente and Lee (2021), Li (2019). *Post-Pandemic:* Weber et al. (2023), Cavallo and Kryvtsov (2024), Chen et al. (2024).

- **Pass-through:**

- Bulow and Pfleiderer (1983), Nakamura and Zerom (2010), Weyl and Fabinger (2013), Mrázová and Neary (2017), Amiti et al. (2019), Butters et al. (2022), Sangani (2025).

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A Case Study: Coffee

Cheapflation and Inflation Inequality Across Food at Home

How much can pass-through in levels explain?

How much do official price indices miss?

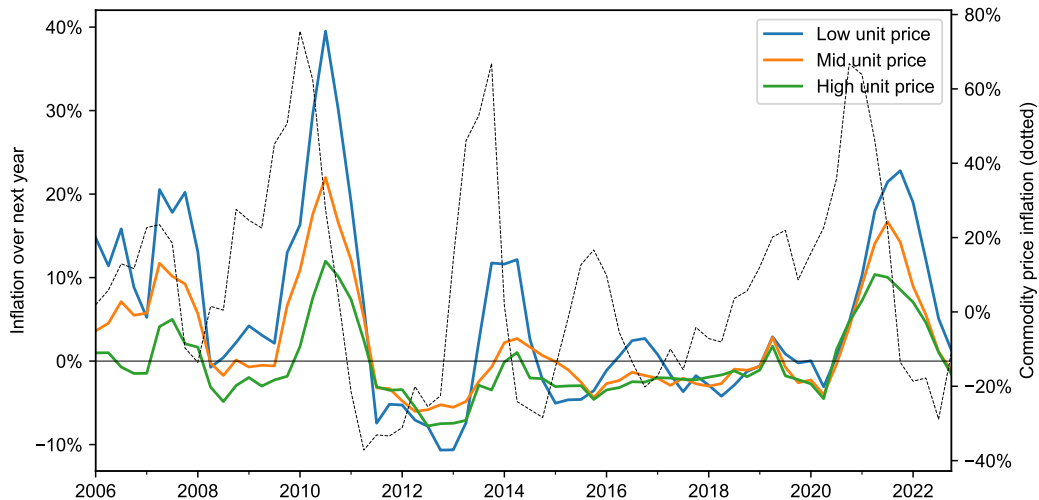
Accounting for substitution and nonhomotheticity

Beyond Food at Home

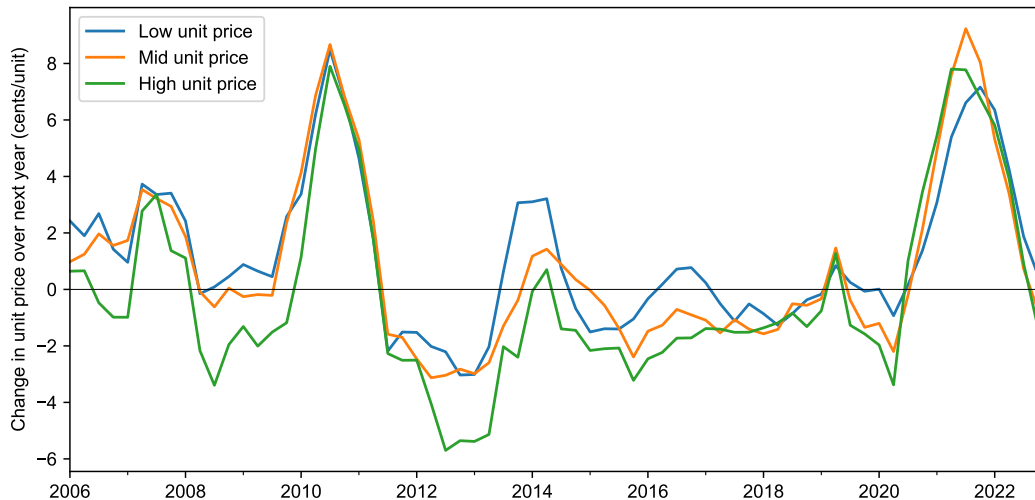
Coffee: A case study

- Roasted coffee as a laboratory, after Nakamura and Zerom (2010), Sangani (2025).
- Primary input, green coffee beans, traded on global commodity markets, with large fluctuations over 2006–2023.
- NielsenIQ Retail Scanner (>30,000 participating U.S. retail stores) for retail prices.
- ① Complete pass-through in levels of input cost increases to prices.
- ② Same absolute price increase $\Rightarrow$ higher inflation for low-priced varieties in category.
- ③ Low-income households buy lower-price varieties, and thus face higher inflation.

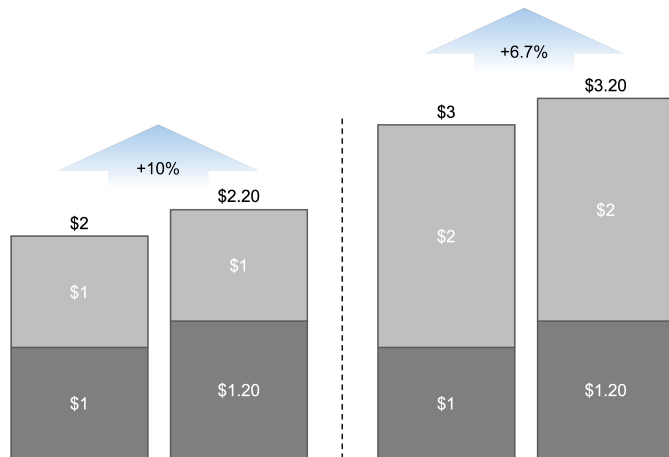
Low-priced products exhibit greater sensitivity of inflation to costs



Differences disappear when measuring price changes in levels



Pass-through in levels and cheapflation: Intuition



- Suppose input price rises from \$1/oz to \$1.20/oz. (+20%).
- L price increases from \$2 to \$2.20. (+10%).
- H price increases from \$3 to \$3.20. (+6.7%).
- Log pass-through:
 $L = 0.5 > H = 0.33$.

Canonical approach to measure pass-through of commodity cost changes

- Specification à la Campa and Goldberg (2005), Nakamura and Zerom (2010), etc.
- Long-run pass-through in levels $\sum_{k=1}^K b_k$ using:

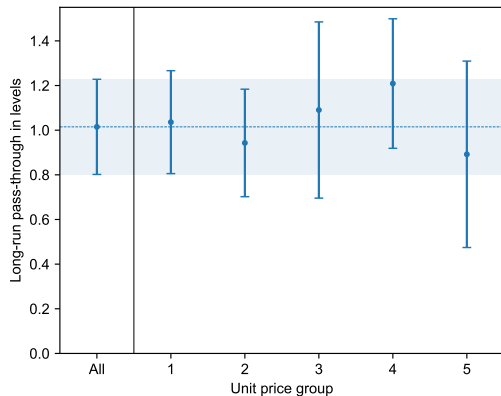
$$\Delta p_{irt} = a_{ir} + \sum_{k=0}^K b_k \Delta c_{t-k} + \sum_{k=1}^4 d_k q_t + \varepsilon_{irt}.$$

- Absorb product-retailer trends a_{ir} and seasonal effects with quarter-of-year dummies q_t .
- Horizon of $K = 6$ quarters, following Nakamura and Zerom (2010).
- Long-run pass-through in logs $\sum_{k=1}^K \beta_k$ from analogous specification,

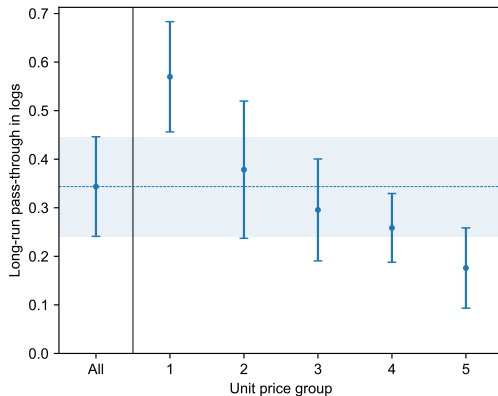
$$\Delta \log p_{irt} = \alpha_{ir} + \sum_{k=0}^K \beta_k \Delta \log c_{t-k} + \sum_{k=1}^4 \delta_k q_t + \varepsilon_{irt}.$$

Complete pass-through in levels $\rightarrow$ heterogeneous log pass-through

- Pass-through in levels is 1.01 (se: 0.11) and uniform across unit price groups.
- Pass-through in logs is incomplete (0.34; se: 0.05) and decreasing with unit price.



(a) Pass-through in levels.



(b) Pass-through in logs.

Robustness

Unit price group	<i>Panel A. Pass-through in levels</i>					
	Baseline	IV: Exchange rates	IV: Weather shocks	Product-quarter FEs	$K = 4$	$K = 8$
1	0.13	0.10	0.25	0.22	0.23	-0.38
2	-0.44	-0.42	-0.17	-0.64	-0.40	0.18
3	0.33	0.52	0.24	-0.02	-0.15	0.06
4	1.06	0.85	-0.32	1.42	1.13	1.30
5	-0.51	-0.36	-0.30	-0.36	-0.52	-0.18

Unit price group	<i>Panel B. Pass-through in logs</i>					
	Baseline	IV: Exchange rates	IV: Weather shocks	Product-quarter FEs	$K = 4$	$K = 8$
1	2.90**	2.21**	1.70*	2.81**	2.57**	2.30**
2	0.39	0.35	0.13	0.25	0.14	1.05
3	-0.64	-0.35	-0.53	-0.88	-0.88	-0.76
4	-1.34	-1.14	-2.04**	-1.02	-0.76	-0.89
5	-2.50**	-2.33**	-2.05**	-2.28**	-2.22**	-2.32**

- Difference in pass-through relative to all products: $(\rho_i - \rho_{\text{all}})/(\text{SE}_i^2 + \text{SE}_{\text{all}}^2)^{1/2}$.
- Result: Uniform pass-through in levels, higher log pass-through for low-price varieties.

Why pass-through in levels?

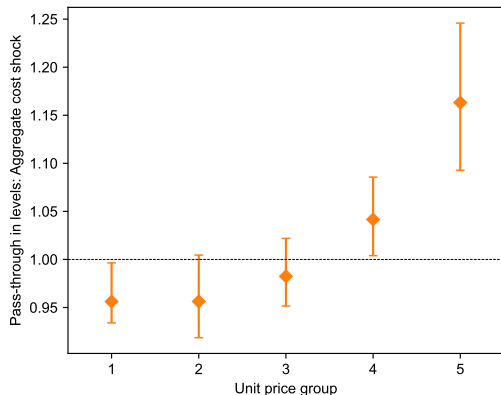
- One explanation: Perfect competition.
 - Differences in prices due to non-commodity input costs. (E.g., Cravino and Levchenko 2017.)
 - At odds with price dispersion, sluggish price adjustment, finite demand elasticities.

Why pass-through in levels?

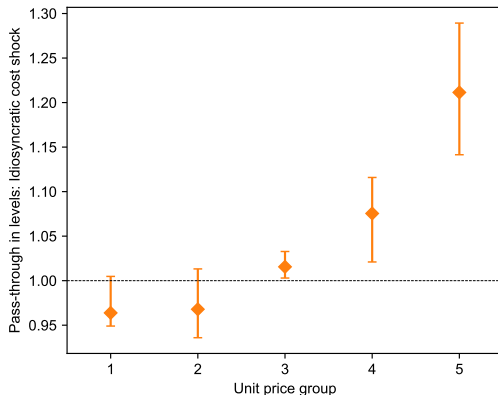
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- Imperfect competition: pass-through depends on how markups adjust to cost changes.
 - No guarantee that firms should exhibit complete pass-through in levels.

No guarantee of pass-through in levels with imperfect competition

- Example: Nakamura and Zerom (2010) demand system for coffee.
- Predicts differences in pass-through of common cost shocks across price distribution.



(a) Common cost shock.



(b) Idiosyncratic cost shock.

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- In fact, homothetic demand predicts complete *log* pass-through of common shocks.
(E.g., CES; Kimball 1995; Atkeson and Burstein 2008; Matsuyama and Ushchev 2017.)

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(E.g., CES; Kimball 1995; Atkeson and Burstein 2008; Matsuyama and Ushchev 2017.)
- One option: **Shift-invariant** demand systems. (Sangani 2025.)
 - Pass-through in levels of common cost shocks (incl. nested/mixed logit, address models).

From cheapflation to inflation inequality

- Inflation rates by income group: Merge NielsenIQ Homescan to Retail Scanner data.
- Sort Homescan panelists into income quintiles.
 - Order by income bin and expenditures/ $\sqrt{\text{household size}}$ within bin. (Handbury 2021).
 - Split into five groups using Nielsen-IQ provided projection weights.
- For each group g , let λ_{igt} = expenditures on product i by group g in quarter t .
- Laspeyres inflation rate using inflation of posted prices for each product i ,

$$\pi_{gt} = \sum_i \frac{\lambda_{igt}}{\sum_j \lambda_{ijt}} \pi_{it}, \quad \text{where} \quad \pi_{it} = p_{it+4}/p_{it} - 1.$$

For match to income groups, define product as a UPC (match 62% of expenditures).

From cheapflation to inflation inequality

- 1 High-income households buy more expensive varieties.

$$\text{LogUnitPrice}_{gt} = \sum_{q=1}^5 \beta_q 1\{g = q\} + \phi_t + \varepsilon_{gt}.$$

- β_q = relative prices of varieties bought by income quintile q .

- 2 High-income face less inflation when costs rise.

$$\text{CoffeeInflation}_{gt} = \sum_{q=1}^5 \beta_q (\text{AvgCoffeeInflation}_t \times 1\{g = q\}) + \alpha_g + \phi_t + \varepsilon_{gt}.$$

- β_q = sensitivity of inflation for income quintile q to overall coffee inflation.

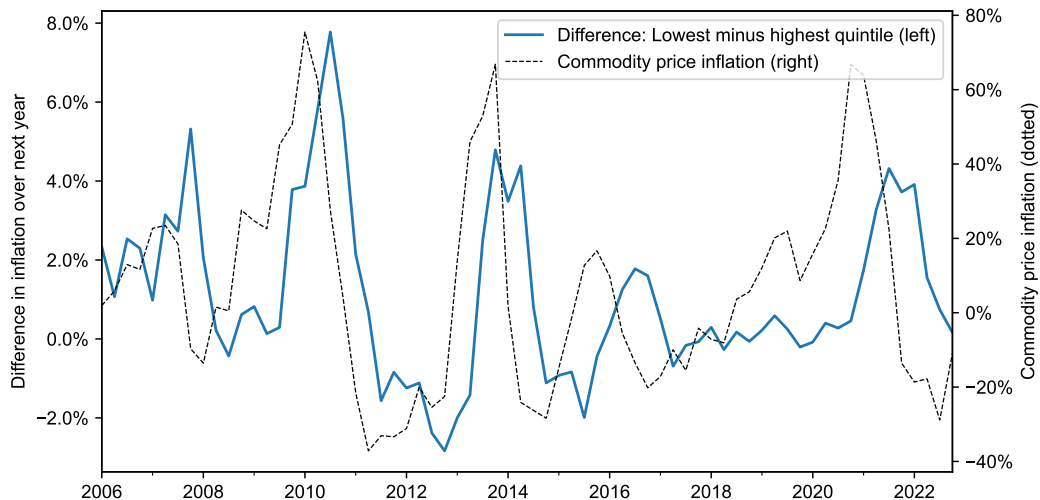
⇒ Construct these measures using NielsenIQ Homescan panelist data.

From cheapflation to inflation inequality

- Low-income buy cheaper varieties $\Rightarrow$ more sensitive to inflation.

<i>Panel A. Prices paid</i>		<i>Panel B. Inflation sensitivity</i>		
	<i>Log unit price</i> (1)		<i>Coffee inflation for income group</i> (OLS)	(IV)
Income quintile 2	0.036** (0.000)	Coffee inflation $\times$ Income quintile 2	-0.039** (0.002)	-0.055** (0.018)
Income quintile 3	0.093** (0.001)	Coffee inflation $\times$ Income quintile 3	-0.096** (0.004)	-0.119** (0.030)
Income quintile 4	0.157** (0.005)	Coffee inflation $\times$ Income quintile 4	-0.155** (0.008)	-0.201** (0.050)
Income quintile 5	0.236** (0.006)	Coffee inflation $\times$ Income quintile 5	-0.270** (0.017)	-0.322** (0.024)
Time FEs	Yes	Time FEs	Yes	Yes
N	360	N	340	340
R^2	0.99	R^2	1.00	0.99
Within R^2	0.93	Within R^2	0.80	0.19

Fluctuations in within-category inflation inequality



- BLS has one ELI for “roasted and instant coffee.” Misses all variation within coffee!

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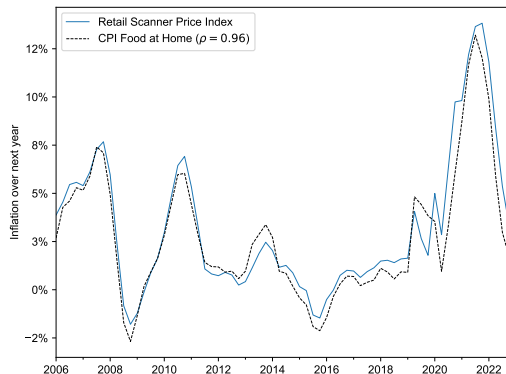
- How much do official price indices miss?

- Accounting for substitution and nonhomotheticity

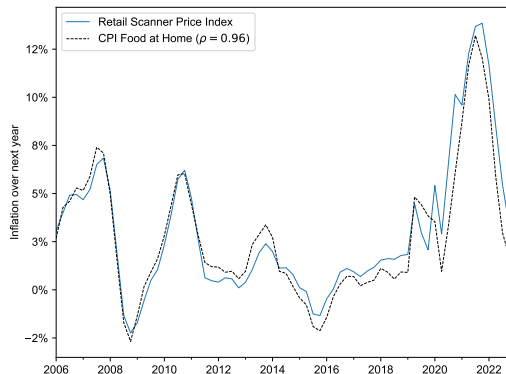
Beyond Food at Home

Does mechanism extend to rest of food at home?

- Replicate BLS food-at-home inflation using UPCs from all NielsenIQ food departments.



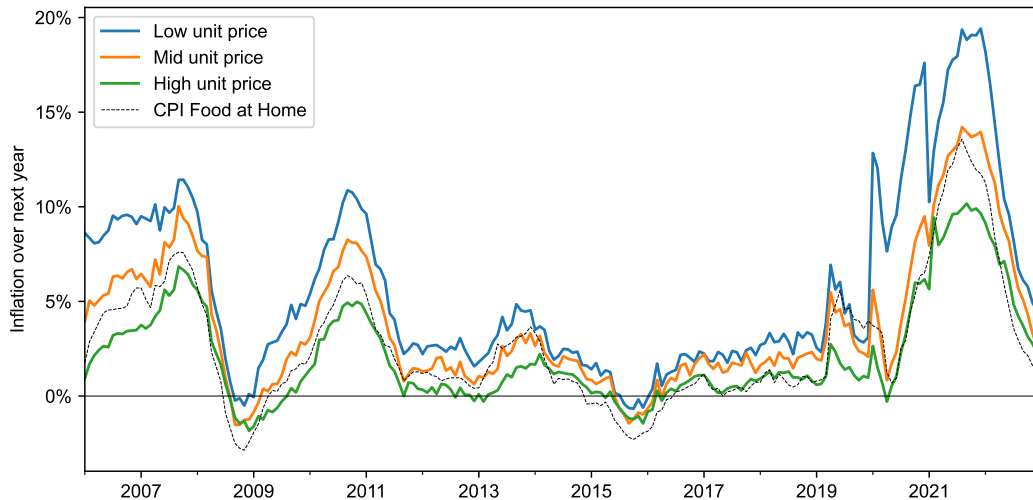
(a) Level of aggregation: Retailer-UPC



(b) Level of aggregation: UPC

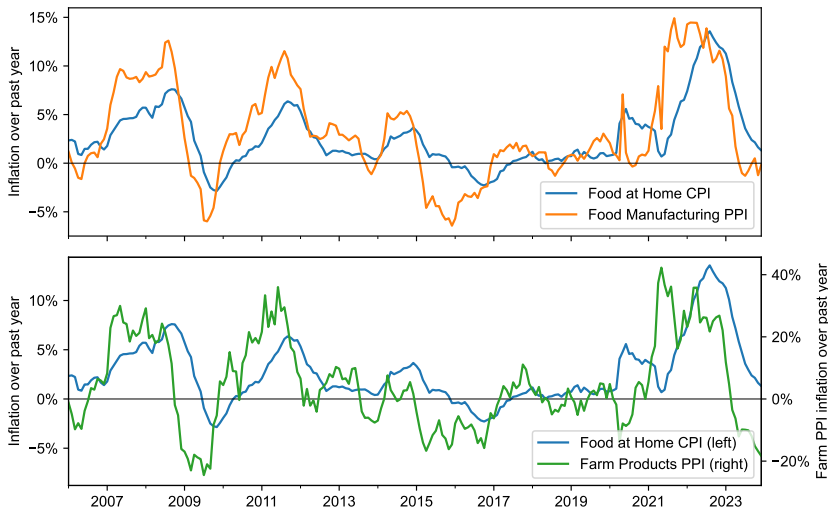
Cycles in food-at-home cheapflation

- Inflation for lower-priced varieties more sensitive to aggregate inflation rate.



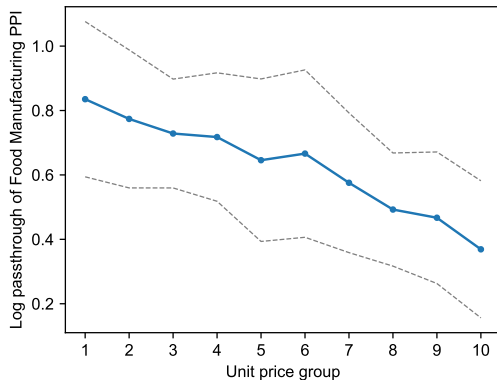
Are cheapflation cycles due to pass-through?

- Log pass-through of Food Mfg. & Farm Products PPI to inflation by unit price group.

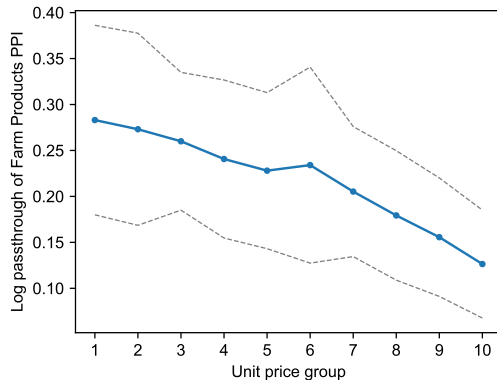


Driven by differential log pass-through of upstream costs

$$\Delta \log p_{it} = \alpha_i + \sum_{k=0}^K \beta_k \Delta \log \text{PPI}_{t-k} + \sum_{k=1}^4 \delta_k q_k + \varepsilon_{it}.$$



(a) Food Manufacturing PPI.



(b) Farm Products PPI.

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Testing for pass-through in levels

- How do we know declining log pass-through is due to complete pass-through in levels?
- **Approach:** Posit simple pricing rule, see if it quantitatively explains patterns.

Testing for pass-through in levels

- How do we know declining log pass-through is due to complete pass-through in levels?
- **Approach**: Posit simple pricing rule, see if it quantitatively explains patterns.
- Suppose each module consists of varieties indexed by i , with unit prices

$$p_i = m + b_i w.$$

- Assumption 1: A unit of each variety contains the same quantity of material input.
- Assumption 2: Demand system generates additive markup priced relative to labor w .
E.g., under generalized logit

$$q_i = \frac{\exp(\delta_i - b_i p_i / w)}{\int_j \exp(\delta_j - b_j p_j / w) dj}.$$

One tractable form of **shift-invariant demand** that explains pass-through in levels.

Testing for pass-through in levels

- Variety i sets price,

$$p_i = m + b_i w.$$

- To a first order, gap between inflation rates for L and H from t to $t + 1$ is

$$\pi_{t+1}^L - \pi_{t+1}^H = \underbrace{\left(\frac{p_t}{p_t^L} - \frac{p_t}{p_t^H} \right)}_{\text{Price gap}} \times \underbrace{(\pi_{t+1}^{\text{all}} - \pi_{t+1}^w)}_{\text{Excess inflation}},$$

- p_t is the average module unit price, and p_t^L, p_t^H are the avg unit prices for L and H .
- $\pi_{t+1}^{\text{all}} - \pi_{t+1}^w$ is difference between module and wage inflation from t to $t + 1$.
- Empirical specification: Indexing modules m and quarters t , we predict $\beta \approx 1$:

$$\text{Cheapflation}_{mt+1} = \beta (\text{PriceGap}_{mt} \times \text{ExcessModuleInflation}_{mt+1}) + \varepsilon_{mt}.$$

Testing for pass-through in levels

- We find $\beta \approx 1$.

	<i>Inflation of lowest-price decile – highest-price decile</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
Excess module inflation $\times$ Price gap	1.183** (0.027)		1.188** (0.025)		1.194** (0.019)	
Module inflation $\times$ Price gap		1.186** (0.025)		1.188** (0.025)		1.194** (0.019)
Wage inflation $\times$ Price gap		-1.657** (0.335)		-1.047** (0.299)		-1.540** (0.453)
Post-2020			-0.067** (0.018)	-0.073** (0.024)		
Time FEs					Yes	Yes
Product Module FEs					Yes	Yes
N	42137	42137	42137	42137	42137	42137
R^2	0.92	0.92	0.92	0.92	0.93	0.93

Testing for pass-through in levels

- We find $\beta \approx 1$. Predictions also hold if we split module vs. wage inflation.

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Testing for pass-through in levels

- We find $\beta \approx 1$. Predictions also hold if we split module vs. wage inflation.
- No excess cheapflation from 2021–2023. Little add'l time variation in cheapflation.

	<i>Inflation of lowest-price decile – highest-price decile</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
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Testing for pass-through in levels: Comparison to alternative models

- ① **Fixed percentage markups:** If $p_i = \mu_i(m + \beta_i w)$, then cheapflation would instead be

$$\pi^L - \pi^H = \left(\frac{\mu_L}{\mu} \frac{p}{p_L} - \frac{\mu_H}{\mu} \frac{p}{p_H} \right) \times (\pi^{\text{all}} - \pi^w).$$

Then, for any $\mu_H > \mu > \mu_L$, we would generically estimate $\beta < 1$ in regression:

$$\pi^L - \pi^H = \underbrace{\beta \left(\frac{p}{p^L} - \frac{p}{p^H} \right)}_{\text{Price gap}} \times \underbrace{(\pi^{\text{all}} - \pi^w)}_{\text{Excess inflation}}.$$

- ② **Consequence of general inflation:** $\beta^{\text{all}} > -\beta^w$.

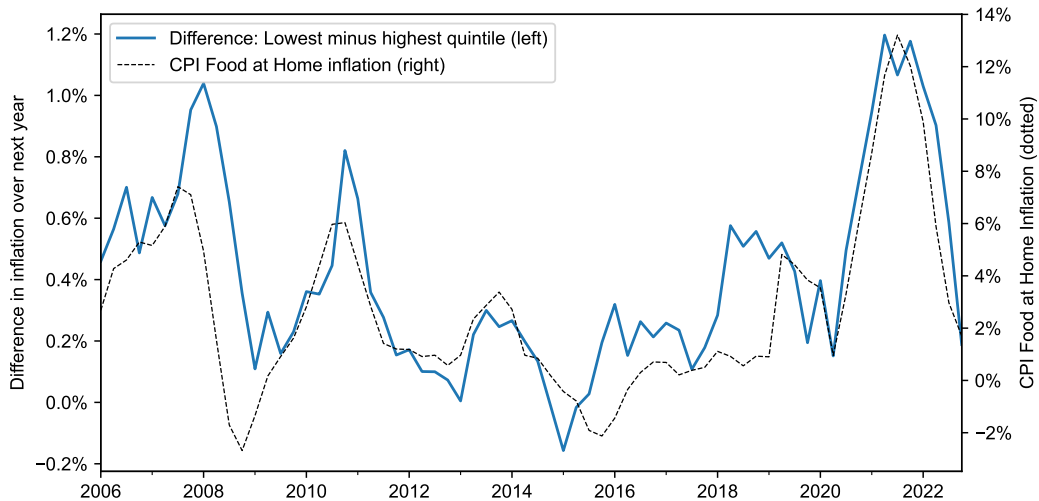
- We cannot reject $\beta^{\text{all}} = -\beta^w$ in any specification ($p = 0.16, 0.65, 0.45$).

From cheapflation to inflation inequality

- Low-income buy cheaper varieties across food-at-home. (Handbury 2021; Sangani 2022).
- $\Rightarrow$ Low-income inflation rates more sensitive to category-level inflation.

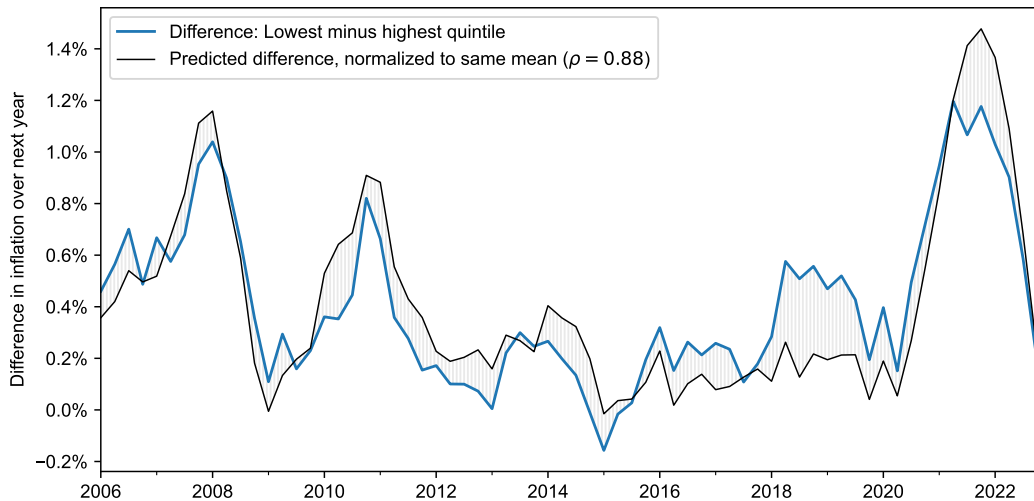
<i>Panel A. Prices paid</i>		<i>Panel B. Inflation sensitivity</i>	
	<i>Log unit price</i> (1)		<i>Module inflation for income group</i> (2)
Income quintile 2	0.012** (0.003)	Avg. module inflation $\times$ Income quintile 2	0.000 (0.003)
Income quintile 3	0.027** (0.004)	Avg. module inflation $\times$ Income quintile 3	−0.008* (0.004)
Income quintile 4	0.049** (0.006)	Avg. module inflation $\times$ Income quintile 4	−0.025** (0.006)
Income quintile 5	0.082** (0.008)	Avg. module inflation $\times$ Income quintile 5	−0.052** (0.010)
Module-Time FEs	Yes	Module-Time FEs	Yes
<i>N</i>	213971	<i>N</i>	201667
<i>R</i> ²	1.00	<i>R</i> ²	0.99
Within <i>R</i> ²	0.19	Within <i>R</i> ²	0.08

Cycles in inflation inequality over entire food-at-home bundle



Cycles in inflation inequality over entire food-at-home bundle

- Predicted gap = diff within-module infl (pass-through in levels) + diff spending shares.



Explaining variation in inflation inequality over time

	<i>Variance of inflation gap (pp²)</i>	<i>Excess inflation inequality relative to full sample average (percentage points)</i>	
	All Years	2008–2011	2021–2023
Actual	0.100	0.122	0.323
After accounting for pass-through in levels	0.043	0.002	0.117
After accounting for pass-through in levels and module expenditure shares	0.030	−0.041	−0.043

- Predicted gap explains 70% of variation in inflation inequality over time.
- Accounts for excess inflation inequality during Great Recession ('08–11) and Post-Pandemic Inflation ('21–23).

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A Case Study: Coffee

Cheapflation and Inflation Inequality Across Food at Home

How much can pass-through in levels explain?

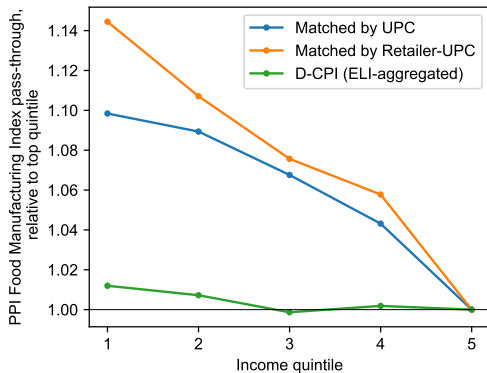
How much do official price indices miss?

Accounting for substitution and nonhomotheticity

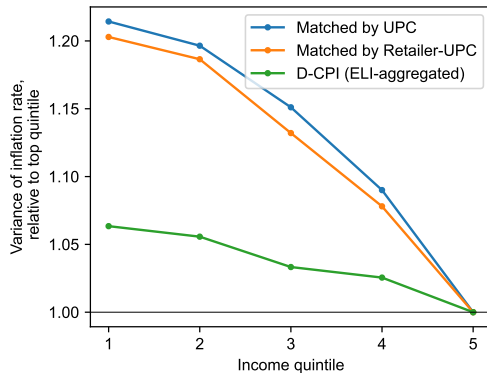
Beyond Food at Home

Official price indices do not capture ↑ volatility, sensitivity for low-income

- Food-at-home inflation for low-income more sensitive to upstream costs, more volatile.
- Inflation measures built on BLS ELI data (e.g., D-CPI) do not capture these effects.



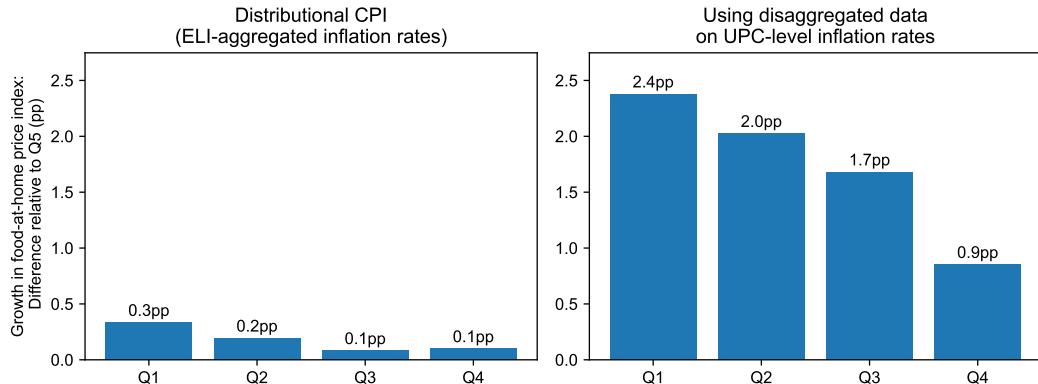
(a) Log pass-through of Food Manufacturing PPI.



(b) Variance of annual inflation rates.

BLS data misses 86% of differential food-at-home inflation over 2021–2023

- Differences in food-at-home price growth by income quintile, 2020Q4 to 2023Q4.
- Average growth was 21pp over this period $\Rightarrow \approx 10\%$ higher for low-income.



Official price indices understate gaps by 70–90%

- “BLS data” column uses most disaggregated official price indices available.
- Aggregation within BLS indices understates differences across income groups.

Measure	<i>Difference: 1st vs. 5th income quintile</i>		
	Scanner data	BLS data	Understatement in BLS data
Pass-through of Food Manufacturing PPI, 2006–2020	9.8%	1.2%	88%
Pass-through of Farm Products PPI, 2006–2020	5.8%	1.6%	72%
Variance of food-at-home inflation rates, 2006–2020	21.4%	6.3%	70%
Growth in food-at-home price index, 2020Q4–2023Q4	2.4pp	0.3pp	86%

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Accounting for substitution and nonhomotheticity

- So far we have used Laspeyres price indices to measure inflation,

$$\underbrace{\Delta \log P_t}_{\text{Change in cost of living}} \approx \sum_i \underbrace{\lambda_{it-1}}_{\text{Initial exp. shares}} \times \underbrace{\Delta \log p_{it}}_{\text{Subsequent price change}} .$$

- Change in cost of living may differ from Laspeyres first-order approx for two reasons:
 - 1 Substitution: Households may substitute to products with smaller price increases.
 - 2 Nonhomotheticity: Households' spending shares may change with real income.

Accounting for substitution and nonhomotheticity

- So far we have used Laspeyres price indices to measure inflation,

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- Change in cost of living may differ from Laspeyres first-order approx for two reasons:

- ① Substitution: Households may substitute to products with smaller price increases.
- ② Nonhomotheticity: Households' spending shares may change with real income.

- Approach:

- ① Törnqvist price index accounts for changes in spending shares.

$$\pi_t^{\text{Törnqvist}} = \prod_i (1 + \pi_{it})^{\frac{1}{2}(\lambda_{it-1} + \lambda_{it})} - 1 .$$

- ② Nonhomothetic second-order adjustment from Jaravel and Lashkari (2024), which uses variation in inflation across hh's with different real consumption.

Accounting for substitution and nonhomotheticity

Income quintile	Panel A. <i>Growth in food-at-home price index from 2020Q4–2023Q4 (pp)</i>			Panel B. <i>Variance of food-at-home inflation rates from 2006–2020 (pp²)</i>		
	Homothetic		Nonhomothetic	Homothetic		Nonhomothetic
	Laspeyres (1)	Törnqvist (2)	(Base=2020Q4) (3)	Laspeyres (4)	Törnqvist (5)	(Base=2006Q1) (6)
1	27.27	25.43	27.01	4.69	4.67	4.89
2	26.92	24.90	26.41	4.62	4.59	4.92
3	26.57	24.52	25.99	4.45	4.43	4.78
4	25.74	23.87	25.27	4.21	4.23	4.55
5	24.89	23.13	24.45	3.86	3.90	4.15
Q1/Q5	1.10	1.10	1.10	1.21	1.20	1.18

- Substitution dampens inflation volatility, $\approx$ proportionately across incomes.
- Nonhomotheticity **amplifies** volatility: households trade down to low-price goods that inflate the fastest.

Why does non-homotheticity amplify volatility / welfare cost?

- In the cross-section, low-income households buy lower-price varieties.
- $\Rightarrow$ As real consumption falls, preferred spending shares shift to cheaper varieties.
- When prices rise, real consumption falls $\Rightarrow$ households want to trade down.
- But, pass-through in levels means households trade down to exactly those varieties that experience the greatest inflation!

Nonhomotheticity adjustment: Robustness to base period

Income quintile	Panel A. <i>Growth in food-at-home price index from 2020Q4–2023Q4 (pp)</i>			Panel B. <i>Variance of food-at-home inflation rates from 2006–2020 (pp²)</i>		
	Baseline	Nonhomothetic		Baseline	Nonhomothetic	
		(<i>b</i> =2020Q4)	(<i>b</i> =2023Q4)		(<i>b</i> =2006Q1)	(<i>b</i> =2020Q4)
	(1)	(2)	(3)	(4)	(5)	(6)
1	27.27	27.01	27.02	4.69	4.89	5.04
2	26.92	26.41	26.35	4.62	4.92	4.84
3	26.57	25.99	25.91	4.45	4.78	4.62
4	25.74	25.27	25.18	4.21	4.55	4.41
5	24.89	24.45	24.42	3.86	4.15	4.09
Q1/Q5	1.10	1.10	1.11	1.21	1.18	1.23

- Similar results using end of period as base.

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Beyond Food at Home

Cheapflation and inflation inequality beyond food at home

- Frequent challenge: How representative is food-at-home of inflation inequality?
- Pass-through in levels should affect inflation inequality in other categories if...
 - Firms have other inputs besides labor,
 - Firms exhibit complete pass-through in levels of these input costs to prices,
 - Consumers purchase differently-priced varieties.
- Two strategies using cross-city variation.
 - 1 Cheapflation across cities.
 - 2 Unequal incidence of inflation across cities with different incomes.
- [\[In paper\]](#): Evidence from vehicle purchases and prices.

Evidence on cheapflation from Cost of Living Index (COLI) data

Category	Products
Grocery	Baby food, Bacon, Bananas, Beer, Chunk light tuna, Coffee, Corn flakes, Eggs, Fresh orange juice, Fried chicken, Frozen corn, Frozen meal, Frozen orange juice, Ground beef, Lettuce, Liquor, Margarine, Parmesan cheese, Peaches, Potato chips, Potatoes, Sausage, Shortening, Soft drink, Sugar, Sweet peas, T-bone steak, Tomatoes, White bread, Whole milk, Wine
Apparel	Boys' jeans, Boys' underwear, Men's denim jeans, Men's dress shirt, Men's slacks, Women's slacks
Other Goods	Aspirin, Board game, Cigarettes, Detergent, Facial tissues, Gasoline, Ibuprofen, Lipitor, Newspaper, Polysporin, Shampoo, Tennis balls, Toothpaste
Services	Beauty salon, Bowling, Commuter fare, Dental visit, Doctor visit, Dry cleaning, Fried chicken (at fast food restaurant), Haircut, Hamburger (at fast food restaurant), Hospital room, Movie ticket, Optometrist visit, Pizza (at restaurant), Tire balancing, Veterinary services, Washer repair
Housing & Utilities	Apartment rent, Electricity (both for all-electric homes and for homes using other types of energy), Home price, Monthly payment, Mortgage rate, Other energy, Telephone bill, Total energy

Evidence on cheapflation from Cost of Living Index (COLI) data

- Cost of Living Index (COLI) price data from 300 cities over 1990–2010.
 - Products chosen to be as standardized as possible across cities.
 - $\Rightarrow$ Prices not necessarily representative of product mix across cities.
- Test for cheapflation using prices of same product i across cities c :

$$\text{Inflation}_{ict+1} = \beta \left(\text{AvgProductInflation}_{it+1} \times \text{RelativePrice}_{ict} \right) + \alpha_{it} + \delta_{ic} + \kappa_{ct} + \varepsilon_{ict},$$

- $\text{RelativePrice}_{ict} = \log p_{ict} - \log \bar{p}_{it}$ is relative price of product i in city c in quarter t .
- Absorb product-quarter, product-city, and city-quarter FEs α_{it} , δ_{ic} , κ_{ct} .
- Lower-priced products more sensitive to inflation (“cheapflation”) implies $\beta < 0$.

Evidence on cheapflation from Cost of Living Index (COLI) data

	<i>Product Inflation in City</i>					
	All (1)	Grocery (2)	Apparel (3)	Other Goods (4)	Services (5)	Housing & Utilities (6)
Avg. Product Infl. $\times$ Relative Price	-1.565** (0.250)	-1.496** (0.329)	-2.440** (0.522)	-1.343** (0.366)	-5.181** (1.996)	-0.586* (0.274)
Product-Quarter FEs	Yes	Yes	Yes	Yes	Yes	Yes
Product-City FEs	Yes	Yes	Yes	Yes	Yes	Yes
City-Quarter FEs	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	1 164 952	516 770	55 111	183 189	262 382	147 500
<i>R</i> ²	0.35	0.38	0.55	0.55	0.27	0.55

- Cheapflation in apparel, other nondurables, and even services.

Evidence on inflation inequality from BLS metro area data

- BLS collects price quotes and reports price indices for 32 metropolitan areas.
- Test for differences in inflation sensitivity, using inflation in MSA m at time t :

$$\text{Inflation}_{mt+1} = \beta (\text{USInflation}_{t+1} \times \text{LogIncome}_{mt}) + \gamma \text{LogIncome}_{mt} + \alpha_m + \delta_t + \varepsilon_{mt},$$

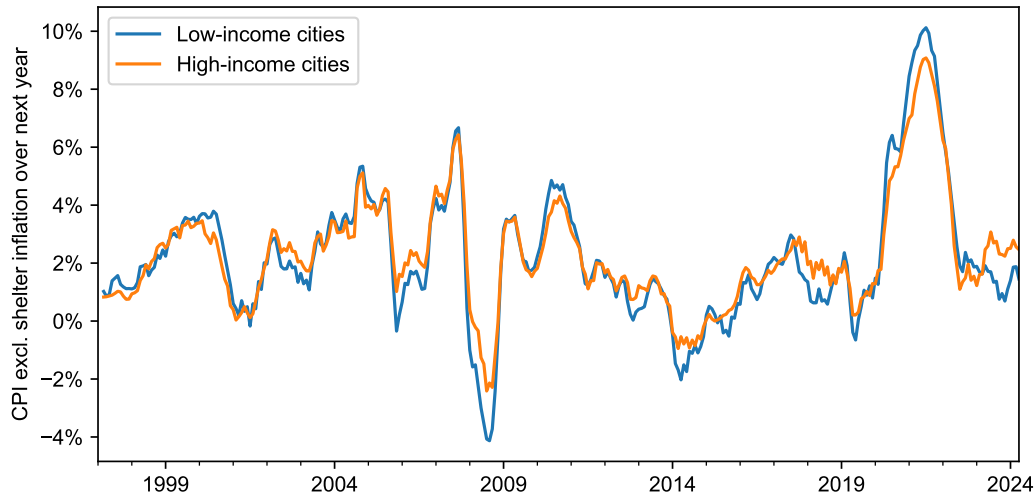
- USInflation_{t+1} is nationwide inflation (U.S. city average) from year t to $t + 1$.
 - LogIncome_{mt} is log of MSA income from BEA.
- $\beta < 0$ implies low-income cities more sensitive to nationwide inflation.

Inflation inequality across U.S. metros

Series BLS Code	<i>Inflation in Metropolitan Area</i>					
	CPI	CPI Ex. Shelter	All Food	Food Away from Home	Goods Ex. Food	Services Ex. Shelter
	SA0 (1)	SA0L2 (2)	SAF1 (3)	SEFV (4)	SACL1 (5)	SASL2RS (6)
U.S. Avg. Inflation $\times$ Log Income	-0.970** (0.085)	-0.452** (0.066)	-0.258** (0.084)	-0.589** (0.242)	-0.227** (0.041)	-0.271* (0.161)
Time FEs	Yes	Yes	Yes	Yes	Yes	Yes
Metropolitan Area FEs	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	3540	3534	3528	3515	3534	3534
<i>R</i> ²	0.85	0.89	0.75	0.42	0.92	0.57

- Higher sensitivity of inflation in lower-income cities.
- Beyond food-at-home, appears in food away from home, other goods, and services.

Inflation of CPI excluding shelter in high- and low-income cities



Differences in expenditure weights across categories explains only half

- BLS publishes expenditure weights across categories for metro areas.
- Construct MSA inflation rates w/ area weights $\times$ national category inflation.
- $\Rightarrow$ Sensitivity of actual MSA inflation rates varies 2x more with income, suggesting differences in within-category product mix and product-level inflation matter.

	<i>Inflation in Metropolitan Area</i>			
	CPI Data		Constructed with Area Weights	
	CPI (1)	CPI Ex. Shelter (2)	CPI (3)	CPI Ex. Shelter (4)
U.S. Avg. Inflation $\times$ Log Income	-0.970** (0.085)	-0.452** (0.066)	-0.208** (0.035)	-0.224** (0.030)
Time FEs	Yes	Yes	Yes	Yes
Metropolitan Area FEs	Yes	Yes	Yes	Yes
<i>N</i>	3540	3534	7452	7452
<i>R</i> ²	0.85	0.89	0.99	0.99

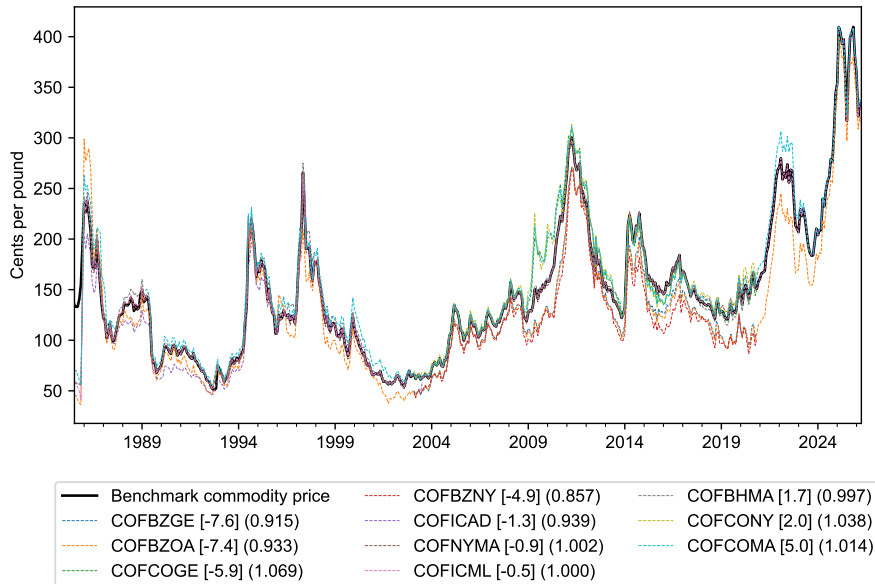
Conclusion

- A new source of fluctuations in cheapflation and inflation inequality.
- Pass-through in levels of cost changes + diff varieties bought by income groups.
- Note: Mechanism applies to any units with different expenditures across varieties.
 - Cities with higher-priced varieties have muted response to national cost shocks.
 - Countries that import high-price varieties have less volatile import price inflation.
 - [In paper](#): Evidence for different coffee inflation across cities / countries.
- Within-category inflation inequality invisible in official statistics due to aggregation.
- Substitution partially mitigates welfare costs, but nonhomotheticity amplifies costs.

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Extra slides

Comparing prices for coffee varieties to benchmark commodity series



Testing for pass-through in levels: Robustness with 5 groups

	<i>Inflation of lowest-price quintile – highest-price quintile</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
Excess module inflation $\times$ Price gap	1.176** (0.027)		1.181** (0.025)		1.188** (0.019)	
Module inflation $\times$ Price gap		1.178** (0.025)		1.181** (0.025)		1.188** (0.019)
Wage inflation $\times$ Price gap		-1.561** (0.299)		-0.926** (0.243)		-1.261** (0.336)
Post-2020			-0.048** (0.011)	-0.056** (0.014)		
Price gap	0.004 (0.010)	0.023* (0.011)	0.012 (0.007)	0.001 (0.006)	0.051** (0.007)	0.054** (0.013)
Time FEs					Yes	Yes
Product Module FEs					Yes	Yes
N	42405	42405	42405	42405	42405	42405
R^2	0.92	0.92	0.93	0.93	0.94	0.94

- Amiti, M., O. Itskhoki, and J. Konings (2019). International shocks, variable markups, and domestic prices. *The Review of Economic Studies* 86(6), 2356–2402.
- Argente, D. and M. Lee (2021). Cost of living inequality during the great recession. *Journal of the European Economic Association* 19(2), 913–952.
- Atkeson, A. and A. Burstein (2008). Pricing-to-market, trade costs, and international relative prices. *American Economic Review* 98(5), 1998–2031.
- Bulow, J. I. and P. Pfleiderer (1983). A note on the effect of cost changes on prices. *Journal of Political Economy* 91(1), 182–185.
- Butters, R. A., D. W. Sacks, and B. Seo (2022). How do national firms respond to local cost shocks? *American Economic Review* 112(5), 1737–1772.
- Campa, J. M. and L. S. Goldberg (2005). Exchange rate pass-through into import prices. *Review of Economics and Statistics* 87(4), 679–690.
- Cavallo, A. (2024). Inflation with covid consumption baskets. *IMF Economic Review* 72(2), 902–917.
- Cavallo, A. and O. Kryvtsov (2024). Price discounts and cheapflation during the post-pandemic inflation surge. *Journal of Monetary Economics* (103644).

- Chen, T., P. Levell, and M. O'Connell (2024). Cheapflation and the rise of inflation inequality. Technical Report 24/36, Institute for Fiscal Studies.
- Cravino, J. and A. A. Levchenko (2017). The distributional consequences of large devaluations. *American Economic Review* 107(11), 3477–3509.
- Del Canto, F., J. Grigsby, E. Qian, and C. Walsh (2025). Are inflationary shocks regressive? A feasible set approach. *Quarterly Journal of Economics* (qjaf028).
- Garner, T., D. S. Johnson, and M. F. Kokoski (1996). An experimental consumer price index for the poor. *Monthly Labor Review* 119(September), 32–42.
- Hagemann, R. P. (1982). The variability of inflation rates across household types. *Journal of Money, Credit and Banking* 14(4), 494–510.
- Handbury, J. (2021). Are poor cities cheap for everyone? non-homotheticity and the cost of living across u.s. cities. *Econometrica* 89(6), 2679–2715.
- Hobijn, B. and D. Lagakos (2005). Inflation inequality in the united states. *Review of Income and Wealth* 51(4), 581–606.
- Hobijn, B., K. Mayer, C. Stennis, and G. Topa (2009). Household inflation experiences in the US: A comprehensive approach. Technical report, Federal Reserve Bank of San Francisco.

- Hochmuth, P., M. Pettersson, and W. Christoffer (2022). A nonhomothetic price index and inflation heterogeneity. Working paper.
- Jaravel, X. (2019). The unequal gains from product innovations: Evidence from the us retail sector. *The Quarterly Journal of Economics* 134(2), 715–783.
- Jaravel, X. (2024). Distributional consumer price indices. Technical report, Working Paper.
- Jaravel, X. and D. Lashkari (2024). Measuring growth in consumer welfare with income-dependent preferences: Nonparametric methods and estimates for the United States. *The Quarterly Journal of Economics* 139(1), 477–532.
- Kaplan, G. and S. Schulhofer-Wohl (2017). Inflation at the household level. *Journal of Monetary Economics* 91, 19–38.
- Kimball, M. S. (1995). The quantitative analytics of the basic neomonetarist model. *Journal of Money, Credit and Banking* 27(4), 1241–77.
- Klick, J. and A. Stockburger (2021). Experimental cpi for lower and higher income households. Technical Report 537, Bureau of Labor Statistics.
- Klick, J. and A. Stockburger (2024, July). Examining u.s. inflation across households grouped by equivalized income. Technical report, U.S. Bureau of Labor Statistics Monthly Labor Review.

- Lan, T., L. Li, and M. Li (2024). Cyclical inequality in the cost of living and implications for monetary policy. Working paper.
- Li, M. (2019). Countercyclical dispersion of price changes: Causes and implications. Working paper.
- Lokshin, M., Z. Sajaia, and I. Torre (2025). Who suffers the most from the cost of living crisis? *Review of Income and Wealth* 71(1), e70009.
- Matsuyama, K. and P. Ushchev (2017). Beyond CES: Three alternative classes of flexible homothetic demand systems. Technical Report 17-109, Global Poverty Research Lab Working Paper.
- Michael, R. (1979). Variation across households in the rate of inflation. *Journal of Money, Credit and Banking* (11), 32–46.
- Mrázová, M. and J. P. Neary (2017). Not so demanding: Demand structure and firm behavior. *American Economic Review* 107(12), 3835–74.
- Nakamura, E. and D. Zerom (2010). Accounting for incomplete pass-through. *The Review of Economic Studies* 77(3), 1192–1230.
- Olivi, A., V. Sterk, and D. Xhani (2024). Optimal monetary policy during a cost-of-living crisis. Working paper.

- Orchard, J. (2022). Non-homothetic demand shifts and inflation inequality. Working paper.
- Pallotti, F., G. Paz-Pardo, J. Slacalek, O. Tristani, and G. L. Violante (2023). Who bears the cost of inflation? Euro Area households and the 2021–2023 shock. Technical Report 31896, National Bureau of Economic Research.
- Sangani, K. (2022). Markups across the income distribution: Measurement and implications. Working paper.
- Sangani, K. (2025). Complete pass-through in levels. Working paper.
- Weber, M., Y. Gorodnichenko, and O. Coibion (2023). The expected, perceived, and realized inflation of us households before and during the covid-19 pandemic. *IMF Economic Review* 72, 326–368.
- Weyl, E. G. and M. Fabinger (2013). Pass-through as an economic tool: Principles of incidence under imperfect competition. *Journal of Political Economy* 121(3), 528–583.